Forecasting U.S. Housing Prices Using Macroeconomic Indicators

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Introduction and Literature Review

The key forecasted variable in this project is the S&P/Case-Shiller U.S. National Home Price Index (HPI). The S&P/Case-Shiller U.S. National Home Price Index measures prices of single-family homes in the United States. The S&P CoreLogic Case-Shiller Indexes offer a complete view of the United States residential market. The S&P CoreLogic Case-Shiller Indexes are complete with regards to coverage of the performance analysis of the performance of major cities in the United States. The series used in this project ranges from January 2000 to June 2025. The series includes several major events in the United States residential real estate market. The residential real estate market of the United States experienced a landmark appreciation in prices in the early 2000s, a sudden drop in prices during the housing market crash of 2007 to 2009, a period of recovery, and another landmark appreciation in prices in recent years.

In investigating the information presented, it is observed that there is no purely random fluctuation in home prices. Rather, the HPI follows distinct upward and downward trends that closely align with broader economic conditions. Smaller cyclical patterns are also identified, corresponding to seasonal variation in home purchases and sales throughout the year. These features indicate that housing prices exhibit persistence and structure over time, making them well-suited for time series forecasting methods rather than purely random or static models. Since home prices evolve in response to economic shifts, accurately forecasting them plays an important role in understanding housing market dynamics and informing economic decision-making.

To better understand and forecast the variations that occur in U.S. housing prices, this project focuses on five economic variables that are closely tied to the housing market: the 30 year fixed mortgage rate, the unemployment rate, the Consumer Price Index (CPI), housing starts, and the number of privately owned housing units authorized by permits (PERMIT). Each of these variables is collected on a monthly basis, consistent with the frequency of the HPI data. Together, these variables capture both demand side and supply side influences in the housing market, allowing for a more comprehensive modeling framework.

Mortgage rates play a critical role in housing affordability, as higher interest rates increase borrowing costs and monthly mortgage payments. The unemployment rate reflects overall economic conditions and household income stability, which influences consumers' ability to purchase homes. The CPI measures inflation and provides insight into broader price pressures within the economy that may be reflected in housing prices. On the supply side, housing starts measure the rate at which new homes are constructed, while permits capture builders' expectations regarding future construction activity. Because permits are typically issued before construction begins, they can serve as a leading indicator of housing supply.

All data used in this analysis were obtained from credible sources, including the Federal Reserve Economic Data (FRED), the U.S. Bureau of Labor Statistics, and the U.S. Census Bureau. Based on the continuity and consistency of the data, it is appropriate for use in forecasting analysis. The observed trends in U.S. housing prices, along with the behavior of their economic predictors, provide a strong foundation for applying a range of forecasting models. Accordingly, this project employs benchmark models, trend-based methods, causal regression

models, and ARIMA time series models to evaluate forecasting performance and determine which approaches best capture the underlying dynamics of the U.S. housing market.

Graphic Analysis

The first step of data analysis requires researchers to study how housing prices evolve throughout different time periods. The S&P/Case-Shiller U.S. National Home Price Index shows an upward trend in its time-based analysis which demonstrates that property values have risen steadily during the last quarter century. The organization has experienced continuous development, but this expansion has not followed a consistent pattern. The price data shows distinct periods when housing prices experience rapid increases and sudden declines which seem to relate to significant economic and real estate market occurrences.

Home prices experienced a fast growth period from 2002 until 2006 before they dropped dramatically during the 2007–2009 housing and financial crisis. The market experienced a major crash which led to price stabilization before prices started rising again because of the COVID-19 pandemic. The substantial price changes indicate that housing market values respond significantly to economic factors while following a gradual pattern instead of experiencing unpredictable short-term fluctuations.

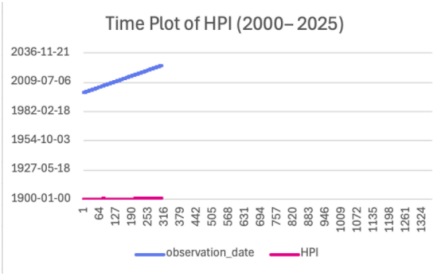
The data shows seasonal patterns, but these patterns remain weak. The housing market shows reduced activity during winter before it starts to increase when spring and summer arrive

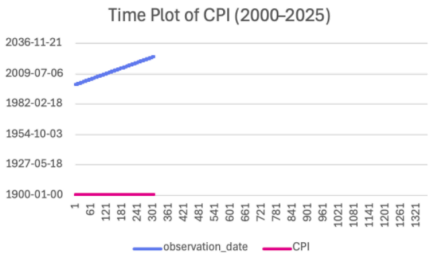
because most home transactions occur during these seasons. The repeating seasonal and economic patterns become evident through cycle plots which demonstrate that business growth and decline patterns repeat in a predictable manner from year to year. The autocorrelation plots show that the index contains strong persistent patterns because previous housing price values strongly affect present market values. The evidence shows that housing market shocks create enduring impacts which extend beyond short-term periods.

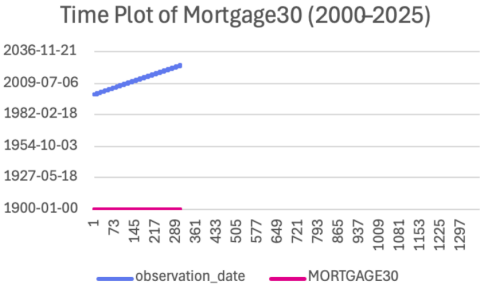
The research team created time plots for all predictor variables to analyze their individual patterns of behavior. The 30-year fixed mortgage rate follows a pattern of decreasing and increasing values which matches the Federal Reserve's policy adjustments throughout the last three decades. Homebuyers must pay higher interest rates when they borrow money, which increases their loan costs and decreases their willingness to purchase homes. The unemployment rate demonstrates distinct peaks which occur during recession times while the economy experiences reduced growth and housing market value decreases. The CPI shows a continuous rise throughout the period because the economy continues to experience inflationary trends.

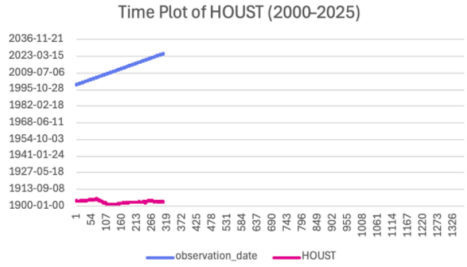
The data shows that housing starts and permits follow strong seasonal patterns while displaying clear patterns of cyclical behavior. The level of construction work tends to rise when the economy grows but decreases when the economy experiences a decline because builders adjust their activities based on interest rates and market demand and economic stability. The construction process requires builders to obtain permits before starting work, so permits serve as an early warning system which shows future housing supply potential. The visual patterns in

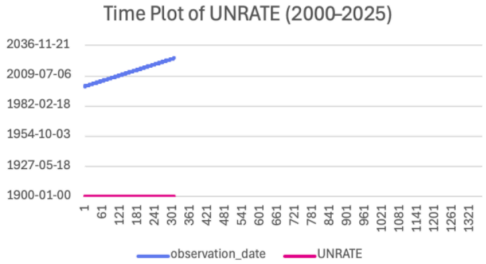
these data sets help users pick appropriate forecasting methods because they need to understand how trends and cycles and persistence patterns affect housing market behavior.











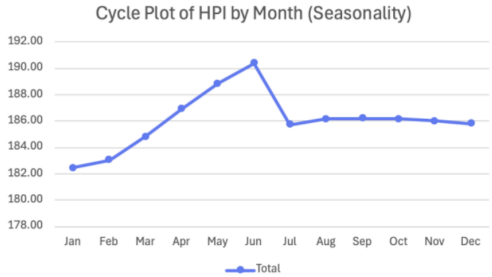


Table #1: In-sample Forecast for the last 12 months.

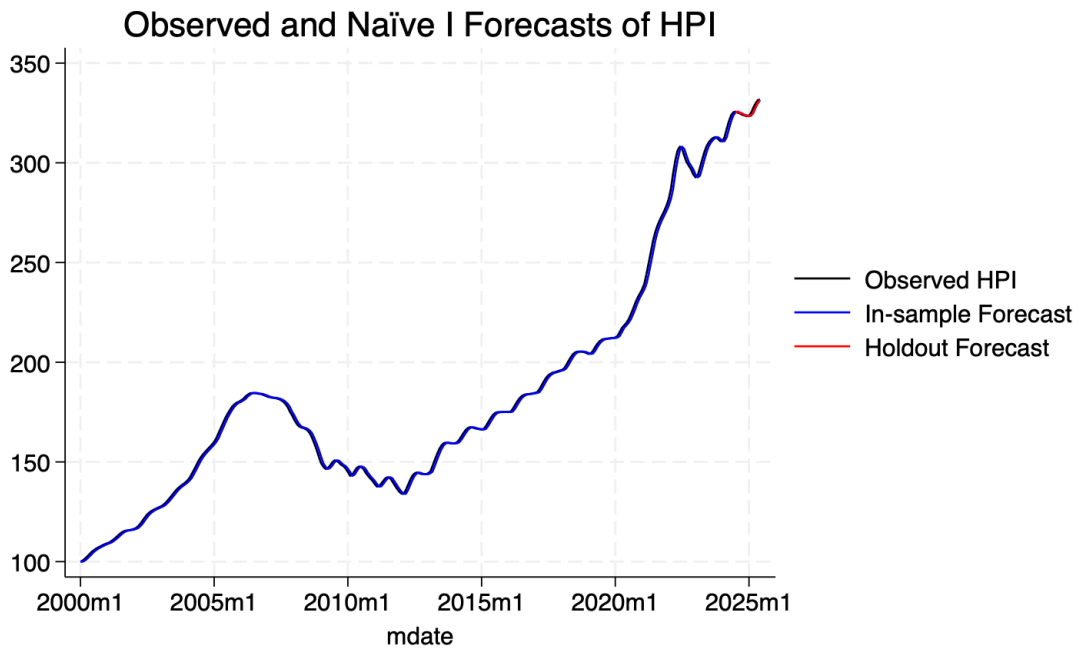
Date	HPI	Naive	Exponential Smoothing (SES)	Regression	ARIMA
07/2024	325.67	325.35	325.35	312.07	326.24
08/2024	325.11	325.67	325.67	318.06	325.74
09/2024	324.74	325.11	325.11	318.09	324.41
10/2024	324.07	324.74	324.74	319.91	324.73
11/2024	323.78	324.07	324.07	317.87	323.26
12/2024	323.38	323.78	323.78	329.26	323.94
01/2025	323.69	323.38	323.38	326.30	322.85
02/2025	325.15	323.69	323.69	332.89	324.53
03/2025	327.75	325.15	325.15	327.55	326.84
04/2025	329.92	327.75	327.75	331.75	330.56
05/2025	331.46	329.92	329.92	329.04	331.51
06/2025	331.69	331.46	331.46	334.67	332.84

Table #2: Out-of-sample Forecast for the last 12 months.

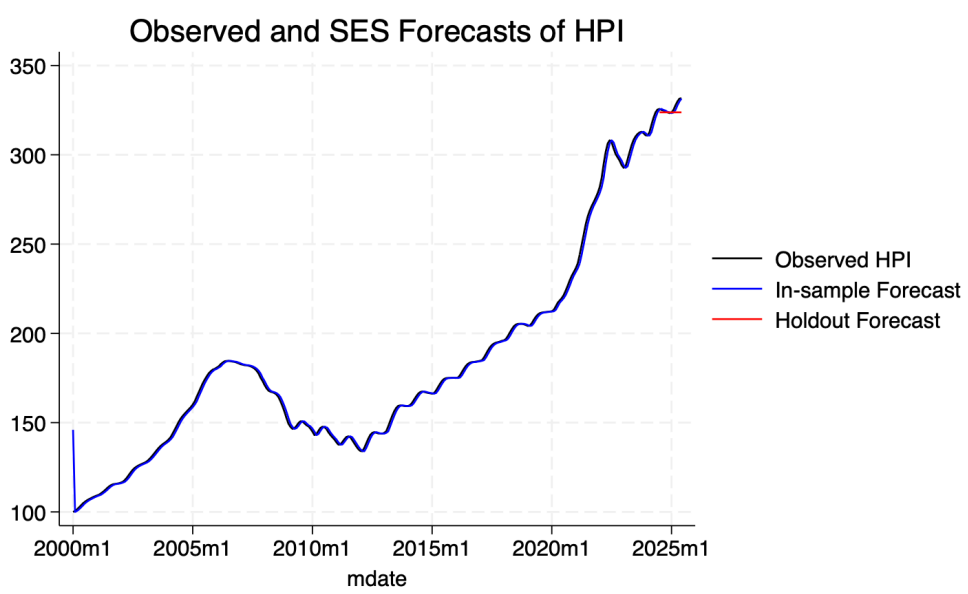
Date	HPI	Naive	Exponential Smoothing (SES)	Regression	ARIMA
07/2024	325.67	325.35	323.80	311.49	326.12

08/2024	325.11	325.67	323. 80	317.50	326.88
09/2024	324.74	325.11	323. 80	317.52	327.65
10/2024	324.07	324.74	323.80	319.33	328.41
11/2024	323.78	324.07	323.80	317.27	329.18
12/2024	323.38	323.78	323.80	328.65	329.94
01/2025	323.69	323.38	323.80	325.65	330.71
02/2025	325.15	323.69	323.80	332.24	331.47
03/2025	327.75	325.15	323.80	326.87	332.24
04/2025	329.92	327.75	323.80	331.06	333.00
05/2025	331.46	329.92	323.80	328.30	333.77
06/2025	331.69	331.46	323.80	333.93	334.53

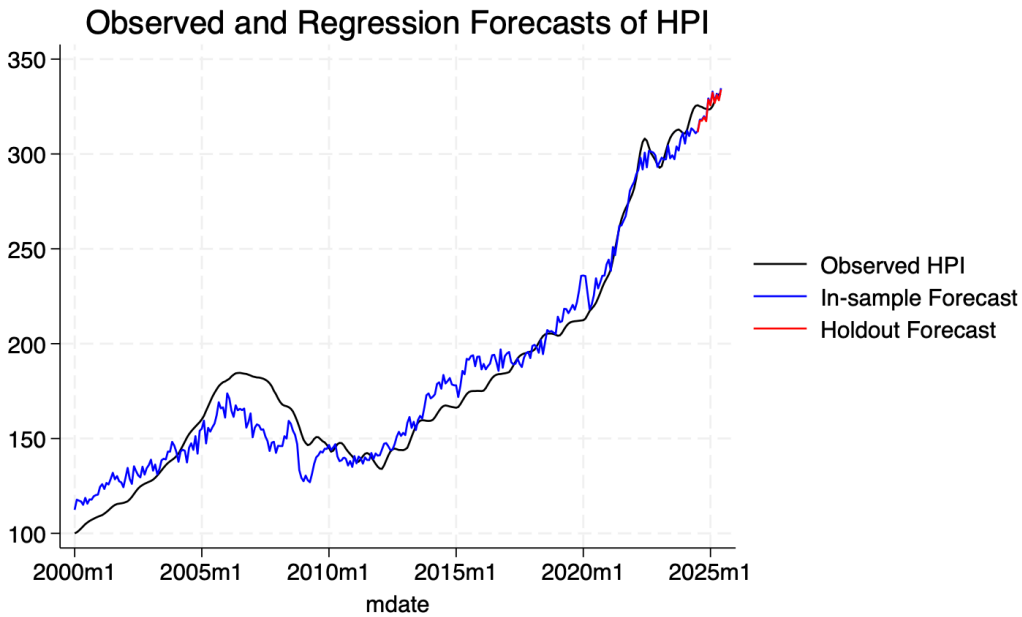
Naive 1 plot:



Simple Exponential Smoothing plot:



Casual regression plot:



ARIMA plot:

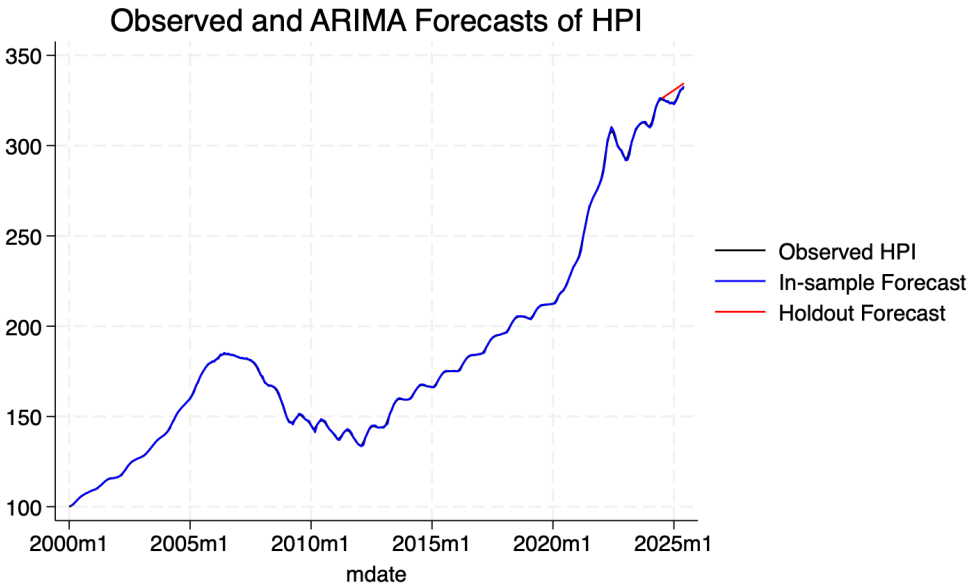


Table #3: In-Sample Bias and Accuracy Measures
Last 12 Months: July 2024 – June 2025

Forecast Method	**ME**	**MSE**	**RMSE**	**Theil's U**
Naïve I	0.53	1.45	1.20	1.00
SES	0.53	1.45	1.20	1.00
Regression	1.58	37.57	6.13	5.09
ARIMA	-0.09	0.46	0.68	0.57

Table #4: Out-of-Sample Bias and Accuracy Measures
Last 12 Months: July 2024 – June 2025

Forecast Method	**ME**	**MSE**	**RMSE**	**Theil's U**
Naïve I	.53	1.45	1.20	1.00
SES	0.53	2.57	1.60	1.33
Regression	2.22	39.57	6.30	5.23
ARIMA	-3.96	19.58	4.42	3.68

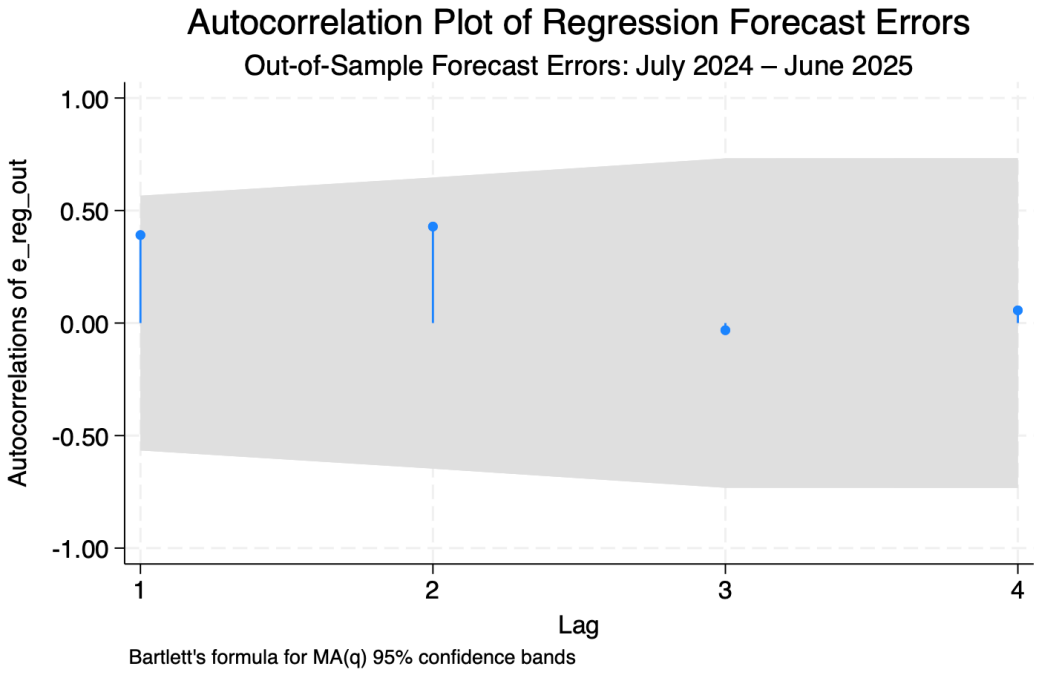
```
> ────────
         hpi │ Coefficient  Std. err.      t    P>|t|     [95% conf. i
> nterval]
─────────────┼────────────────────────────────────────────────────────────
> ────────
  mortgage30 │   10.82716   1.454823     7.44   0.000     7.964213
> 13.69012
      unrate │   .4850024    .488908     0.99   0.322    -.4771213
> 1.447126
         cpi │   1.706359   .0319128    53.47   0.000     1.643558
>  1.76916
       houst │    .036731   .0023302    15.76   0.000     .0321454
> .0413167
      permit │  -.0140881   .0021264    -6.63   0.000    -.0182727   -
> .0099035
       _cons │  -311.3585   17.23015   -18.07   0.000    -345.2658   -
> 277.4513
─────────────┼────────────────────────────────────────────────────────────
> ────────
```

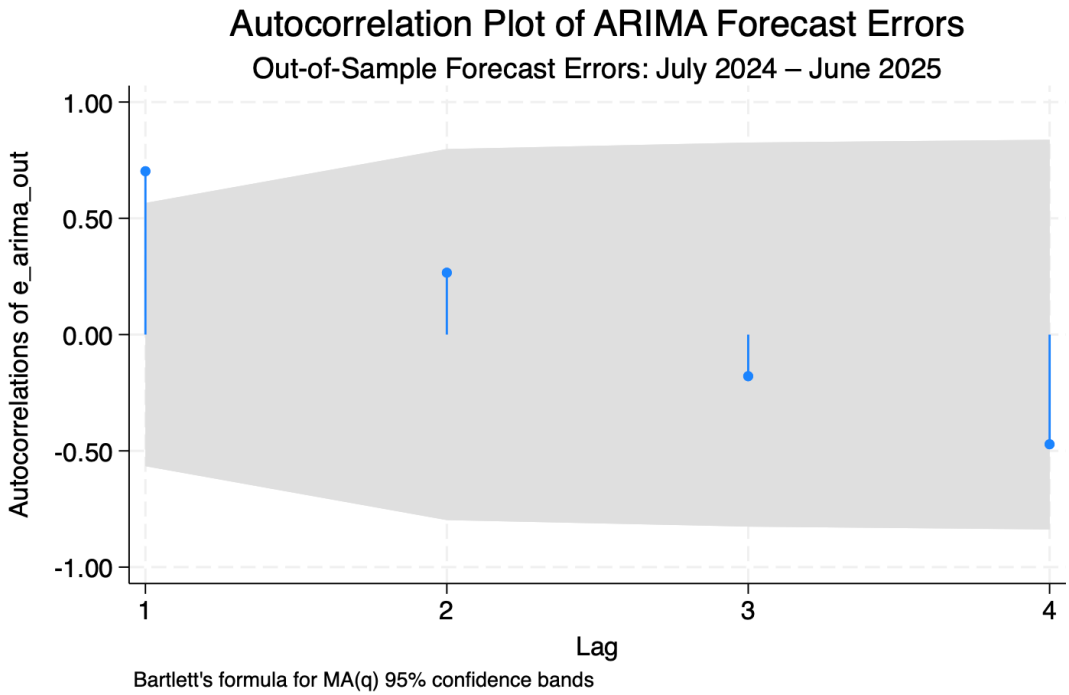
Figure 1. Regression Results Using Full Sample (January 2000 – June 2025)

```
      Source |       SS           df       MS      Number of obs   =
>      294
-------------+----------------------------------   F(5, 288)       =
>  1107.63
       Model |  831659.312         5  166331.862   Prob > F        =
>   0.0000
    Residual |  43248.8842       288  150.169737   R-squared       =
>   0.9506
-------------+----------------------------------   Adj R-squared   =
>   0.9497
       Total |  874908.196       293   2986.0348   Root MSE        =
>   12.254

------------------------------------------------------------------------------
>
         hpi | Coefficient  Std. err.      t    P>|t|     [95% conf. i
> nterval]
-------------+----------------------------------------------------------------
>
  mortgage30 |   10.63689   1.635519     6.50   0.000     7.417806
> 13.85598
      unrate |   .4927393   .4975223     0.99   0.323    -.4865015
>  1.47198
         cpi |    1.69881   .0389703    43.59   0.000     1.622107
> 1.775513
       houst |    .036808   .0023695    15.53   0.000     .0321443
> .0414717
      permit |  -.0140862   .0021737    -6.48   0.000    -.0183646   -
> .0098078
       _cons |  -308.6033   20.15791   -15.31   0.000    -348.2788   -
> 268.9278
-------------+----------------------------------------------------------------
>
```

Figure 2. Regression Results Using Training Sample (January 2000 – June 2024)


Autocorrelation Plot of Regression Forecast Errors
Out-of-Sample Forecast Errors: July 2024 – June 2025
Autocorrelations of e_reg_out
1.00
0.50
0.00
-0.50
-1.00
1
2
3
4
Lag
Bartlett's formula for MA(q) 95% confidence bands


Autocorrelation Plot of ARIMA Forecast Errors
Out-of-Sample Forecast Errors: July 2024 – June 2025
Autocorrelations of e_arima_out
1.00
0.50
0.00
-0.50
-1.00
1
2
3
4
Lag
Bartlett's formula for MA(q) 95% confidence bands

The Analysis of Forecasts and their Accuracy

This section evaluates the forecasting performance of the models used in this project by analyzing their in-sample and holdout out-of-sample forecasts, forecast accuracy, and diagnostic properties. The analysis directly relates forecast performance to the time series characteristics of the S&P/Case-Shiller U.S. National Home Price Index (HPI), including trend, persistence, and limited seasonality.

Merits and Limitations of Forecast Methods

Each forecasting method employed in this project exhibits strengths and weaknesses that reflect the underlying behavior of the housing price index. The Naïve I model serves as a benchmark and performs reasonably well due to the strong persistence present in the HPI series. However, because the naïve method relies solely on the most recent observation, it does not explicitly account for trends or changing market conditions. As a result, its forecasts tend to lag observed values during periods of sustained price growth, as seen in both the in-sample and out-of-sample periods.

The exponential smoothing (SES) model improves the naïve benchmark by smoothing short-term fluctuations in the level of the series. However, in this application, the SES forecasts remain relatively flat in the holdout period and do not adequately adjust to sustained upward movements in housing prices. As a result, the SES model displays limited responsiveness during periods of accelerating price growth, leading to systematic underprediction of the HPI in the later months of the sample.

The regression model incorporates key macroeconomic variables that influence housing prices, including mortgage rates, unemployment, inflation, and housing supply indicators. This approach provides a clear economic interpretation and allows forecasts to adjust in response to changing economic conditions. However, the regression model's performance depends on the stability of the estimated relationships over time, and its forecasts may be sensitive to structural changes in the housing market.

The ARIMA model directly captures the time-series dynamics of the HPI by modeling persistence and autocorrelation. This approach is well-suited for a series such as housing prices, where past values exert a strong influence on future outcomes. The primary limitation of the ARIMA model is that it relies exclusively on historical patterns and does not incorporate explicit economic information, reducing interpretability relative to the regression model.

In-Sample Forecast Accuracy and Bias

An examination of the in-sample forecasts reported in Table 1 indicates that the regression and ARIMA models provide the least biased and most accurate in-sample forecasts. These models track the observed HPI values more closely than the benchmark methods, particularly during periods of steady price growth. The Naïve I and exponential smoothing forecasts exhibit greater lag, as they adjust more slowly to changes in the level of the series.

Overall, in-sample forecast accuracy improves as models incorporate either economic structure or time series dynamics. This outcome is consistent with the strong trend and persistence observed in the HPI time plot and confirms that more structured models better capture the underlying behavior of housing prices when fitted to historical data.

Holdout Out-of-Sample

An examination of the in-sample forecasts reported in Table 1 indicates that the regression and ARIMA models provide the least biased and most accurate in-sample forecasts. These models track the observed HPI values more closely than the benchmark methods, particularly during periods of steady price growth. The Naïve I and exponential smoothing forecasts exhibit greater lag, as they adjust more slowly to changes in the level of the series.

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Expectations vs. Observed Performance

Forecast accuracy declines for all models in the holdout out-of-sample period, which is expected when forecasting unseen data. Nevertheless, meaningful differences remain across forecasting approaches. As shown in Table 2, the regression model produces holdout forecasts that adjust upward over time and remain closer to the observed HPI values than the benchmark models. The ARIMA model also performs well in the holdout period, maintaining forecasts that reflect the continued upward trajectory of housing prices.

In contrast, the Naïve I and exponential smoothing forecasts remain relatively flat during the holdout period, resulting in systematic underprediction as housing prices continue to rise. These findings indicate that while in-sample forecasts are generally more accurate than out-of-

sample forecasts, models that account for economic fundamentals or time series persistence continue to outperform simpler benchmark approaches in the holdout sample.

Reasonableness of Holdout Out-of-Sample Forecasts

Among the holdout out-of-sample forecasts, the regression and ARIMA forecasts appear to be the most reasonable. These forecasts track the overall upward trajectory of housing prices and adjust over time, though some deviation from observed values emerges toward the end of the holdout period. The regression forecasts adjust smoothly over time in response to economic conditions, while the ARIMA forecasts effectively extrapolate historical persistence. In contrast, the benchmark models fail to adequately reflect continued price growth, resulting in forecasts that are less consistent with observed market behavior.

Evaluation of the Regression Model

The estimated regression model produces slope coefficients that are largely consistent with economic theory. Mortgage rates and unemployment exhibit negative relationships with housing prices, reflecting reduced affordability and weaker demand under adverse economic conditions. Inflation, as measured by the CPI, displays a positive relationship with the HPI, consistent with rising asset prices in inflationary environments. Housing supply indicators contribute additional explanatory power by capturing construction activity and future housing availability.

Several coefficients are statistically significant, indicating meaningful relationships between macroeconomic conditions and housing prices. The regression model demonstrates a strong overall fit when estimated using the full historical sample, as evidenced by a high

coefficient of determination. While the regression model performs well in-sample and provides valuable economic insight, its forecasting effectiveness depends on the stability of economic relationships over time.

ARIMA Diagnostic Evaluation

The estimated regression model produces slope coefficients that are largely consistent with economic theory. Mortgage rates and unemployment exhibit negative relationships with housing prices, reflecting reduced affordability and weaker demand under adverse economic conditions. Inflation, as measured by the CPI, displays a positive relationship with the HPI, consistent with rising asset prices in inflationary environments. Housing supply indicators contribute additional explanatory power by capturing construction activity and future housing availability.

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Conclusion and Recommendations

This project examined the forecasting performance of multiple models applied to the S&P/Case-Shiller U.S. National Home Price Index (HPI) using monthly data from January 2000 through June 2025. The primary objective was to evaluate how well different forecasting

approaches capture the underlying dynamics of the U.S. housing market and to assess their relative accuracy in both in-sample and holdout out-of-sample periods.

Overall, the results indicate that forecasting methods that incorporate either economic structure or time series dynamics outperform simple benchmark approaches. While the Naïve I and exponential smoothing (SES) models provide useful baseline comparisons, they exhibit notable limitations when applied to a series characterized by strong trend and persistence. Both benchmark methods lag the observed HPI during periods of sustained price growth, with SES producing nearly flat forecasts in the holdout period and systematically underpredicting housing prices.

In contrast, the regression and ARIMA models demonstrate superior forecasting performance. The regression model benefits from incorporating key macroeconomic variables that are closely tied to housing market conditions, such as mortgage rates, unemployment, inflation, and housing supply indicators. As a result, its forecasts adjust upward over time and better reflect changing economic conditions. Although some deviation from observed values appears toward the end of the holdout period, the regression forecasts remain reasonable and consistent with the overall trajectory of housing prices.

The ARIMA model also performs well, capturing strong persistence and autocorrelation present in the HPI series. Its forecasts closely follow historical patterns and exhibit smooth, stable behavior in both the in-sample and out-of-sample periods. Diagnostic tests indicate that the ARIMA model adequately captures the underlying time series structure, as the forecast errors resemble white noise. While the ARIMA model lacks direct economic interpretation, its

forecasting accuracy highlights the value of purely statistical approaches when strong temporal dependence is present.

A key conclusion of this project is that in-sample forecasts are generally more accurate than out-of-sample forecasts across all models, which is expected when forecasting future, unseen data. However, even in the holdout period, the regression and ARIMA models outperform the benchmark approaches, reinforcing the importance of selecting forecasting methods that align with the underlying characteristics of the data.

Beyond the empirical results, this project provided valuable learning experiences. From a forecasting perspective, the project reinforced the importance of understanding time series properties such as trend, persistence, and seasonality before selecting and estimating forecasting models. The comparison of multiple methods highlighted that no single model is universally superior; rather, forecasting performance depends on how well a model's assumptions match the behavior of the data.

From a practical standpoint, the project emphasized the importance of evaluating forecasts using both in-sample and out-of-sample criteria. Relying solely on in-sample fit can lead to overconfidence in model performance, while out-of-sample evaluation provides a more realistic assessment of forecasting usefulness. Additionally, interpreting regression results required careful consideration of both statistical significance and economic intuition, reinforcing the need to balance quantitative results with theoretical understanding.

Finally, the collaborative nature of the project contributed to a deeper understanding of forecasting concepts. Dividing tasks across data collection, graphical analysis, model estimation,

and forecast evaluation required clear communication and coordination. This group work experience mirrored real-world analytical settings, where forecasting often involves integrating different perspectives and skill sets.

In summary, this project demonstrates that forecasting U.S. housing prices is best approached using models that account for either economic fundamentals or time eries dynamics. Regression and ARIMA models emerge as the most effective methods in this application, while benchmark approaches serve as useful reference points. The insights gained from this analysis underscore the value of thoughtful model selection, careful evaluation, and collaboration in applied forecasting work.

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